

Srividya Bandari

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SUMMARY

Software Engineer with 4+ years of experience building distributed backend services, event-driven workflows, real-time streaming systems, and ML-assisted decision pipelines. Strong background in .NET Core, Java, Kafka, Pulsar, Flink, Redis, PostgreSQL, AWS/Azure, observability, and production reliability. Delivered measurable impact across high-volume systems, including 35% workflow latency reduction, 70% VIN decode p95 improvement, 45% faster issue detection, and stronger release safety.

WORK EXPERIENCE

Software Engineer

July 2023 – Present

Openlane — Tech: .NET Core, Apache Pulsar, Redis, PostgreSQL, Terraform, Honeycomb

Austin, TX

- Owned core services in a distributed .NET Core inspection workflow platform processing 300K+ daily vehicle inspections and 3.5M+ validation, integration, audit, and observability events/day across inspection, inventory, VIN intelligence, and downstream systems.
- Designed Apache Pulsar-based event propagation with replay-safe consumers, correlation IDs, schema-versioned payloads, and failure-isolated retry paths, reducing workflow latency by 35% while improving real-time state consistency.
- Built ML-assisted inspection decision pipelines by enriching workflow events with VIN intelligence, vehicle metadata, inspection signals, and historical state-transition features, generating risk scores used by rule engines and manual-review queues to detect anomalous transitions, duplicate processing signals, and inconsistent vehicle attributes.
- Implemented idempotent event consumers with deduplication keys, exponential backoff, dead-letter handling, and vendor-failure isolation, reducing duplicate processing risk and preventing invalid inspection state transitions during downstream degradation.
- Optimized high-volume VIN decoding and reference-data lookups using Redis-backed caching, TTL-based invalidation, and request coalescing, improving p95 latency by 70% while reducing repeated third-party vendor calls.
- Led production rollout with Terraform-managed feature flags, staged enablement, canary deployments, rollback paths, and Honeycomb distributed tracing, reducing MTTD by 45% and improving release safety across critical inspection workflows.

Research Software Engineer Intern

Jan 2022 – July 2022

Mayo Clinic — Tech: Python, PyTorch, TensorFlow, Multi-view CNNs, SageMaker, Datadog

Phoenix, AZ

- Designed a modular multi-view CNN training and inference pipeline in PyTorch, fusing image embeddings with structured metadata features to support reproducible model development across medical-imaging datasets.
- Improved model performance through class-imbalance handling, loss-function tuning, threshold calibration, and 1K+ hard-case error analysis, raising accuracy from 87% to 95% while reducing false-negative rate from 13% to 5%.
- Added PyTorch hooks, tensor-level tracing, and Datadog dashboards for feature extraction, inference latency, and failure diagnostics, reducing MTTR from 2 days to 4 hours and increasing trace-event coverage from 40% to 100%.

Software Engineer

Jan 2020 – Aug 2021

Infosys — Tech: Java, Go, Spring Boot, Kafka, Flink, Redis, RocksDB, AWS

Hyderabad, India

- Built a real-time DDoS detection pipeline using Kafka, stateful Flink, and Java on Kubernetes, ingesting 2M+ network flow records/sec and cutting detect-to-mitigate latency from 25s to under 8s.
- Implemented a Go/Java mitigation orchestration service over gRPC to push staged BGP Flowspec, RTBH, and ACL actions with audit logging, operator approval, and rollback safeguards, automating 90%+ of approved mitigations.
- Developed adaptive baselining and heavy-hitter detection with Count-Min Sketch, EWMA, RocksDB-backed Flink state, and Redis caching, reducing compute and memory footprint by 40% with no loss in precision or recall.
- Drove observability and reliability with Prometheus, Grafana, OpenTelemetry, Splunk, SLO dashboards, and chaos drills, achieving 99.97% availability and reducing mean time to mitigation from 11s to 5s.
- Implemented secure multi-tenant rollout workflows on AWS using EKS, S3, CloudWatch, IAM, KMS, Terraform, Argo Rollouts, and feature flags, enabling tenant-level isolation, staged releases, and safer rollback across client environments.

PROJECTS

RAG-Powered Document Q&A System | Python, FAISS, FastAPI, Docker, Vector Search

- Built a FastAPI-based RAG service for PDF ingestion, semantic chunking, FAISS retrieval, context ranking, and grounded Q&A; added retrieval-quality checks that reduced irrelevant responses by 40% versus baseline generation.

Multi-Threaded File System Simulator | C++, POSIX APIs, Read-Write Locks, Journaling, Concurrency

- Built a POSIX-style file system simulator in C++ with concurrent reads/writes, block allocation, inode-style metadata indexing, journaling, and crash recovery; improved throughput by 70% over a single-threaded baseline under contention.

TECHNICAL SKILLS

Languages: C#, Java, Python, SQL, Go, JavaScript, TypeScript

Backend: .NET Core, ASP.NET Core, Spring Boot, REST, gRPC, Microservices, Distributed Systems

ML/Data: PyTorch, TensorFlow, SageMaker, FAISS, Snowflake

Streaming & Messaging: Apache Pulsar, Kafka, RabbitMQ, Event-Driven Architecture

Databases & Caching: PostgreSQL, SQL Server, Redis, MongoDB, Elasticsearch/OpenSearch

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Jenkins

Observability: Honeycomb, Datadog, Grafana, Prometheus, Splunk, OpenTelemetry

EDUCATION

Arizona State University – Master of Science, Computer Engineering

Aug 2021 – May 2023

Osmania University – Bachelor of Engineering, Electrical and Electronics Engineering

Aug 2016 – Jan 2020